

High-speed Serial Interface

Lect. 13 – Digital PLL

(최광천)

May

iversity

일	월	화	수	목	금	토
			1 Lect. 10	2	3	4
5	6 Lect. 11	7	8 Lect. 12	9	10	11
12	13 Lect. 13 Digital PLL	14	15 Lect. 14 CDR1	16	17	18
19	20 Lect. 15 CDR2	21	22 Lect. 16 CDR3	23	24	25
26	27 SP	28	29 SP	30	31	

June

diversity

일	월	화	수	목	금	토
						1
2	3 SP	4	5 Quiz 2	6	7	8
9	10 Lab 2	11	12	13	14	15
16	17	18	19	20	21	22
	← Final Exam Period →					
23 30	24	25	26	27	28	29

- **Evaluation**

- Quiz: $3 \times 10 = 30$ → Quiz $2 \times 15 = 30$
- Lab Reports: $2 \times 15 = 30$
- Student Presentation: $2 \times 15 = 30$
- Class Participation: 10

● Paper List for SP#2

(5/27)

“A CMOS Self-Calibrating Frequency Synthesizer”, IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 35, NO. 10, OCTOBER 2000 (문정욱)

“A low noise PLL design by loop bandwidth optimization” IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 35, NO. 6, JUNE 2000 (유병민)

(5/29)

”Jitter Optimization Based on PLL Design Parameters” IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 37, NO. 11, NOVEMBER 2002 (반유진)

“Analysis and modeling of bang bang clock and data recovery circuits” IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 39, NO. 9, SEPTEMBER 2004 (박정현)

● Paper List for SP#2

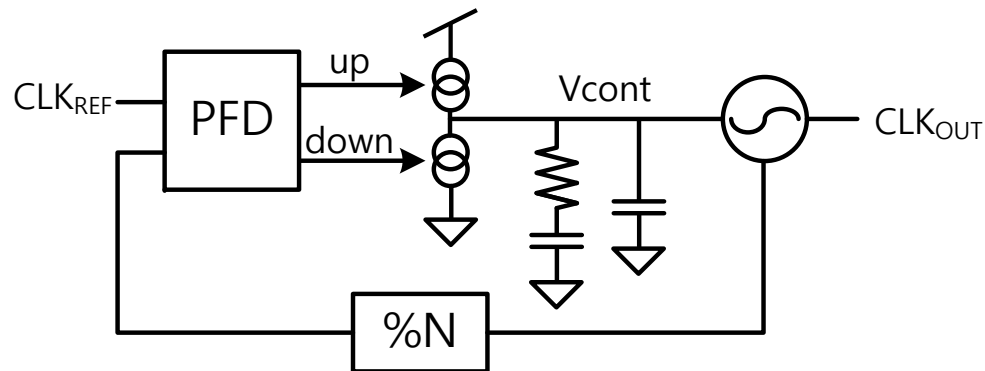
(6/3)

“A 10Gbps CMOS clock and data recovery circuit with a half rate linear phase detector” IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 36, NO. 5, MAY 2001 (김민형)

“A 40Gbps Clock and Data Recovery Circuit in 018um CMOS Technology” IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 38, NO. 12, DECEMBER 2003 (권대현)

- **What is ADPLL?**
 - Limit of conventional PLLs
 - What is ADPLL?
 - Advantages/disadvantages of ADPLL
- **Building blocks**
 - Digital loop filter
 - Time to digital converter
 - Digitally-controlled oscillator

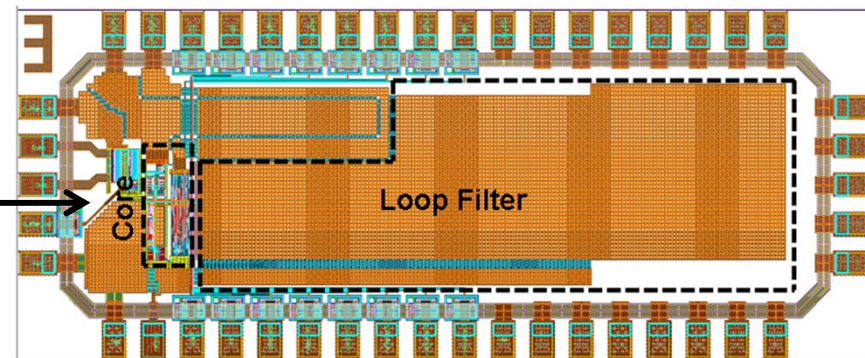
- Conventional CP-based PLL (analog PLL)



- Problem 1

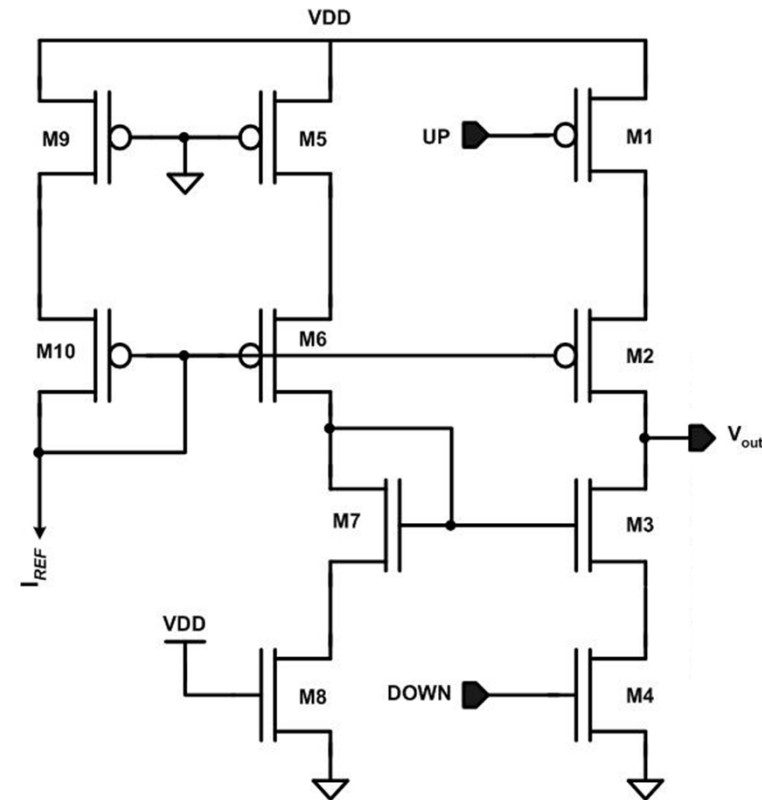
- Layout of typical CPPLL →

PFD + CP
+ VCO



- Loop filter passive components are hard to be scaled down even though CMOS technologies are advanced.
- External loop filter causes additional noises and requires more pads, PCB area.

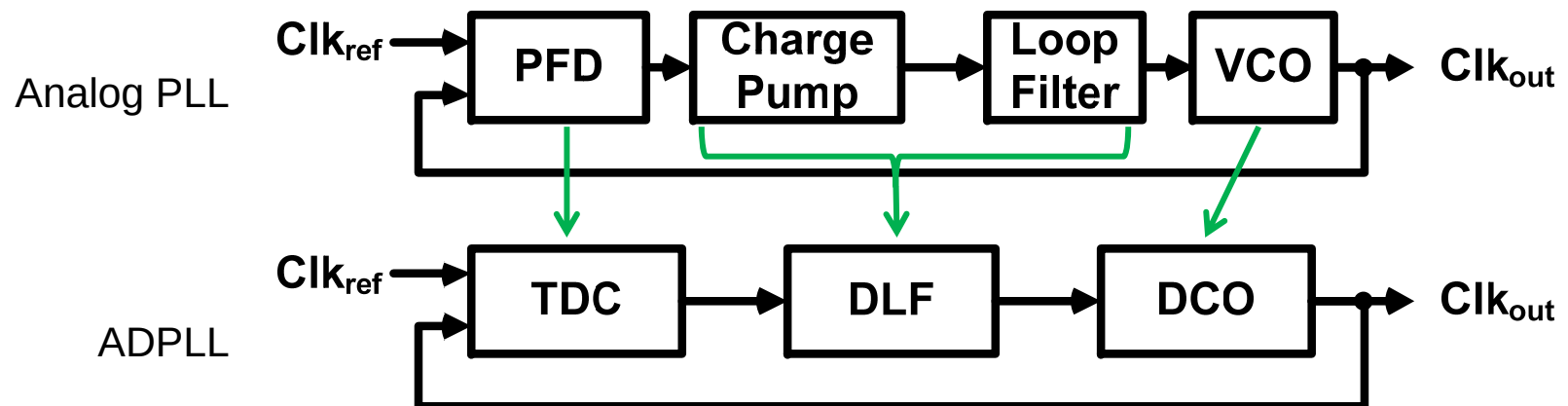
- Problem 2
 - Schematic of conventional CP →



- VDD & V_{th} are scaled down → CP leakage, mismatch degrade PLL jitter performance.
- Problem 3
 - Leakage current through LF capacitor is serious below 90-nm devices.

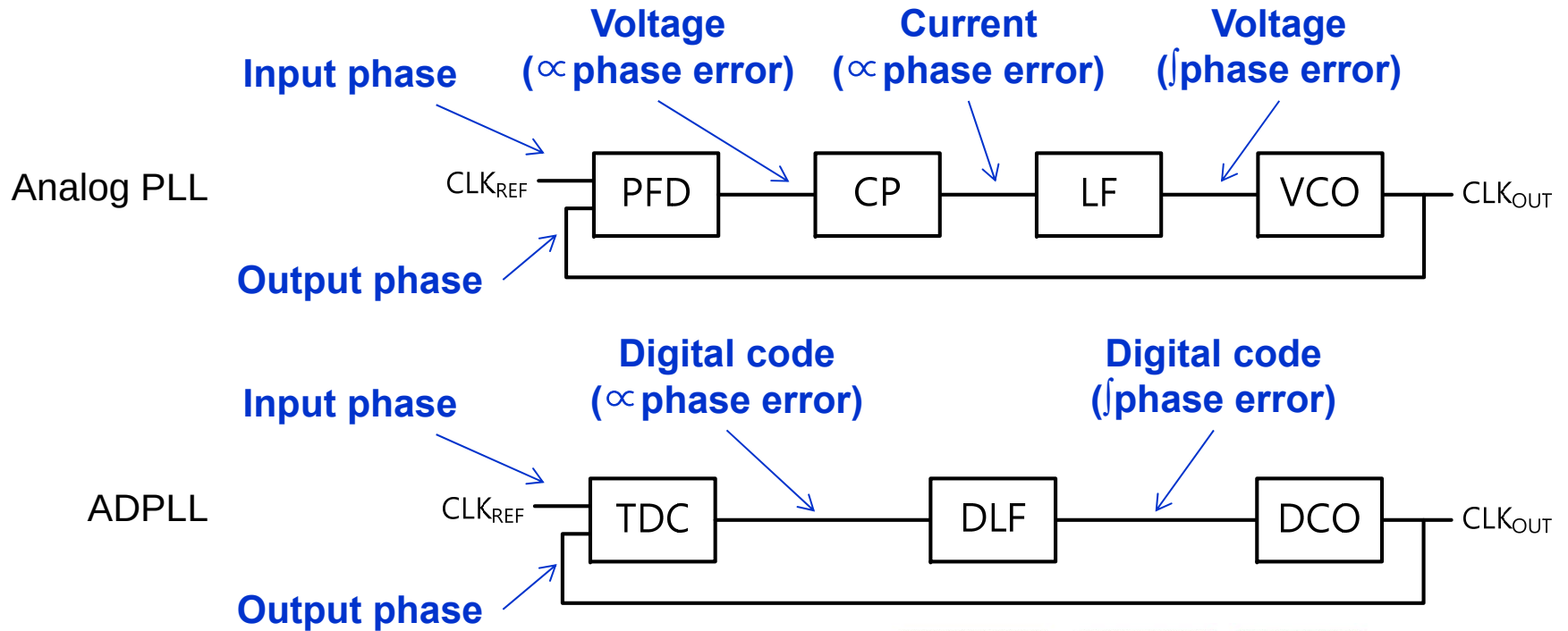
What is ADPLL?

- What is ADPLL?
 - CP + LF $\rightarrow$ Digital loop filter.
 - For this, PFD $\rightarrow$ TDC (Time-to-Digital Converter), VCO $\rightarrow$ DCO (Digitally-Controlled Oscillator)
 - TDC produces digital code which is proportional to the input phase/frequency error.
 - DCO produces output clock of which frequency is proportional to the input digital code.

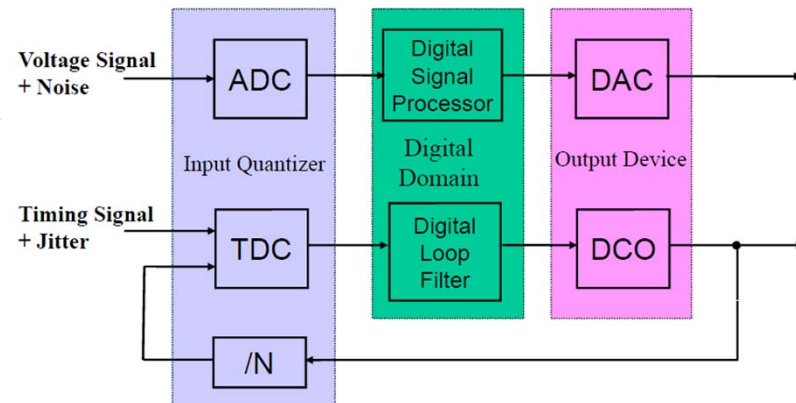


Yonsei university

- Signals in CPPLL & ADPLL



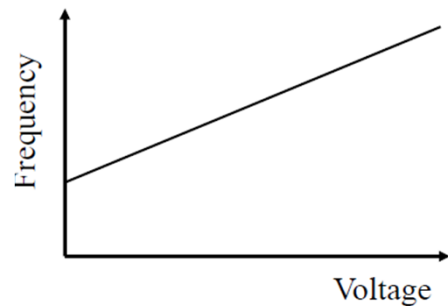
- Similar with typical DSP system →



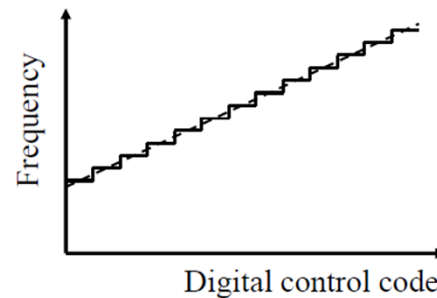
- Advantages
 - Large passive components are not required.
 - CP & LF non-ideality problem can be reduced.
 - Supply voltage can be decreased easily.
 - PVT variation can be covered more easily.
 - Digital loop filter is unaffected by PVT variation.
 - Filter coefficient can be compensated easily.
 - Portability
 - Synthesizable
 - Cell-based ADPLL: All building blocks are synthesizable.
 - Frequency/phase information can be processed more flexibly.
 - Fast locking with initialization possible.

- Disadvantages

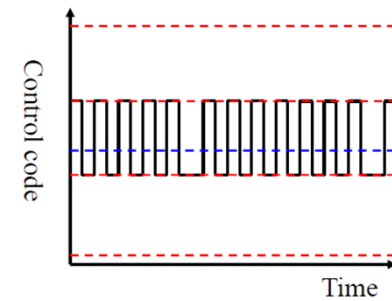
- DCO quantization noise $\rightarrow$ DCO resolution should be fine.



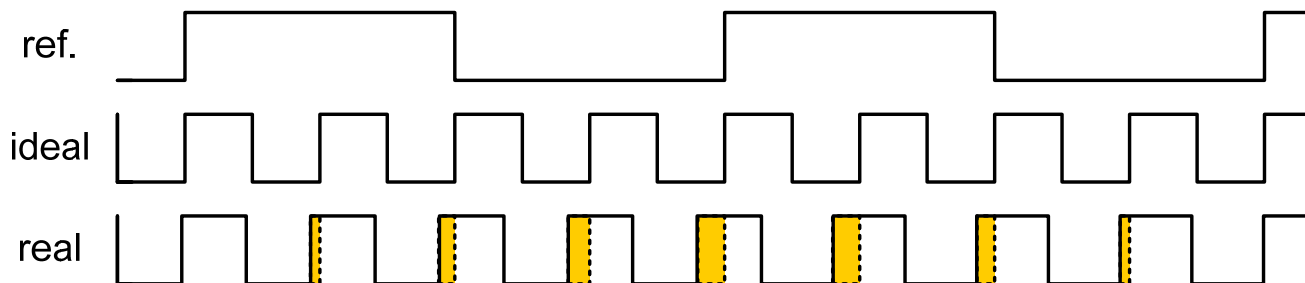
VCO: continuous frequency



DCO: quantized frequency



DCO control code in locked state
 $\rightarrow$ statistic jitter



- DCO trade off: resolution, frequency range, complexity

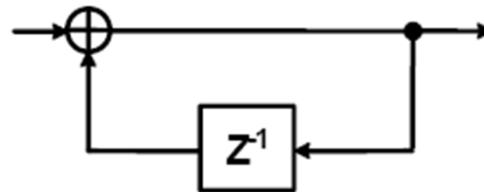
- Disadvantages
 - TDC quantization noise → TDC resolution should be fine.
 - TDC trade off: resolution, dynamic range, maximum operating speed

- Digital loop filter
- Time to Digital Converter
 - Bang-Bang
 - Linear
 - Logarithmic
- DCO
 - Explicit DAC + VCO/CCO
 - Embedded DAC + Oscillator

- Digital loop filter
 - PVT independent filter
 - Easy to design: simple Z-domain transformation from S-domain filter
 - Easy to select coefficients of digital filters
 - One-time calibration
 - Adaptive on the fly
 - Easy expansion to higher order filter

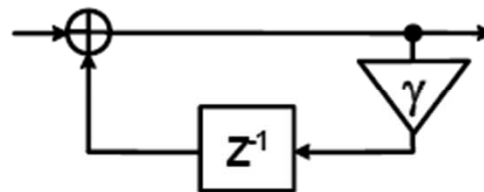
- Some simplest form of Z-domain transfer functions

Integrator



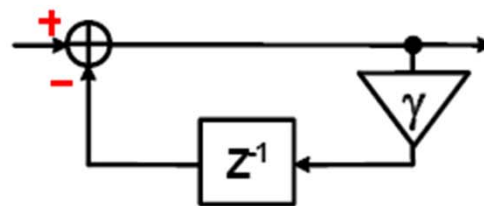
$$H(z) = \frac{1}{1 - z^{-1}}$$

Low pass



$$H(z) = \frac{1}{1 - \gamma z^{-1}}$$

High pass

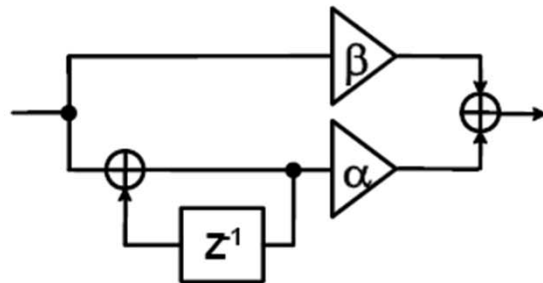


$$H(z) = \frac{1}{1 + \gamma z^{-1}}$$

- Filter design from famous continuous-time filter



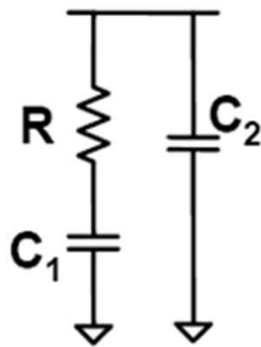
$$H(s) = \left(R + \frac{1}{sC} \right) = \frac{(s/\omega_z + 1)}{sC}$$
$$\omega_z = \frac{1}{RC}$$



$$H(z) = \left(\beta + \frac{\alpha}{1 - z^{-1}} \right)$$

Proportional path Integral path

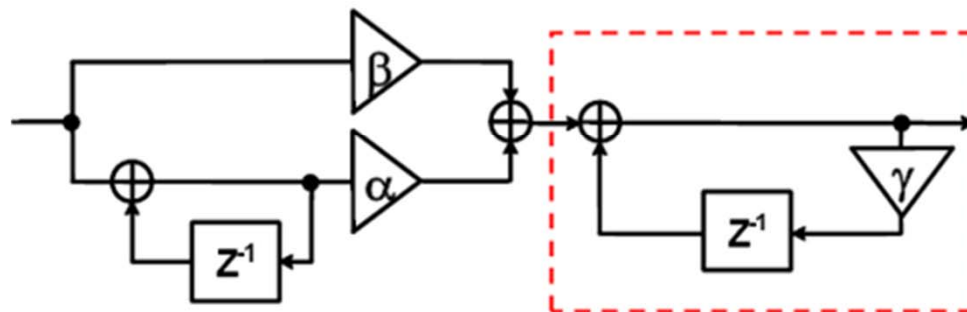
- Filter design from famous continuous-time filter



$$H(s) = \frac{(s/\omega_z + 1)}{(C_1 + C_2)s} \bullet \frac{1}{(s/\omega_p + 1)}$$

$$\omega_z = \frac{1}{RC_1}, \omega_p = \frac{C_1 + C_2}{RC_1C_2}$$

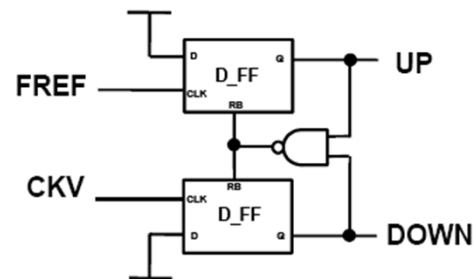
← Additional
Low pass filter



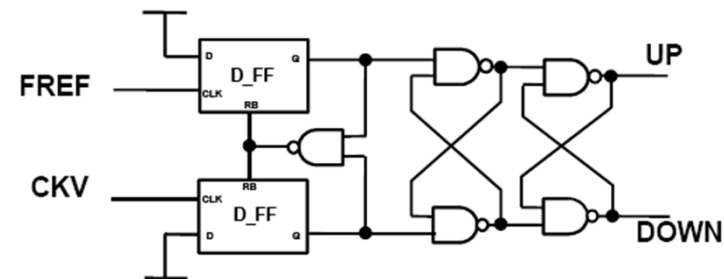
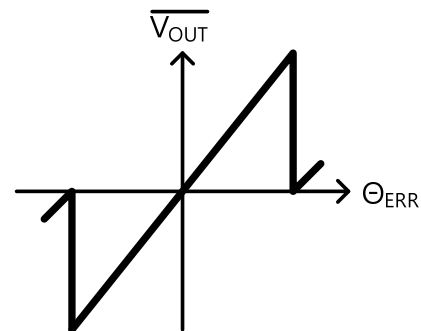
Additional
Low pass filter

$$H(z) = \left(\beta + \frac{\alpha}{1 - z^{-1}} \right) \bullet \frac{1}{1 - \gamma z^{-1}}$$

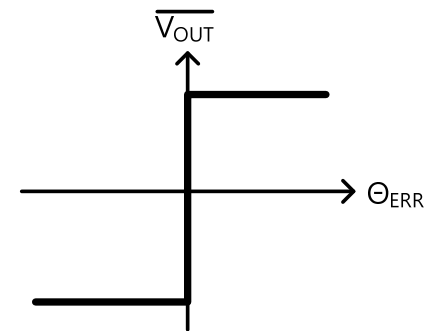
- Bang-bang TDC
 - Bang-Bang type PFD



Conventional PFD

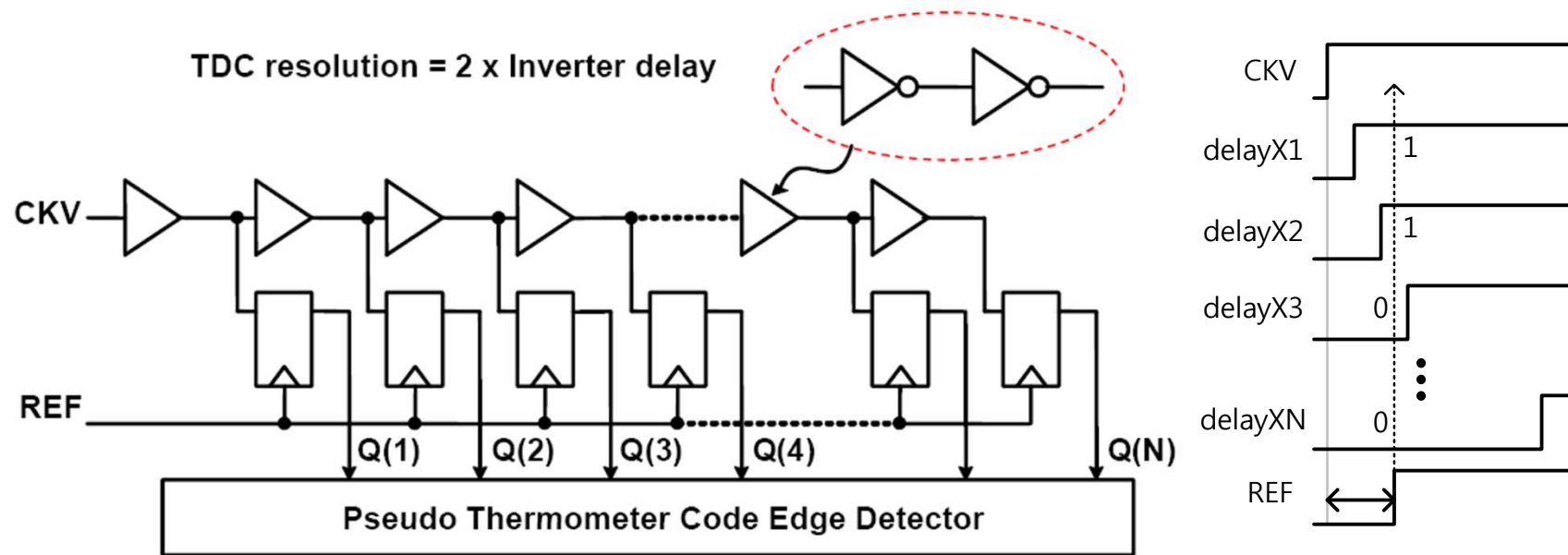


BB-PFD



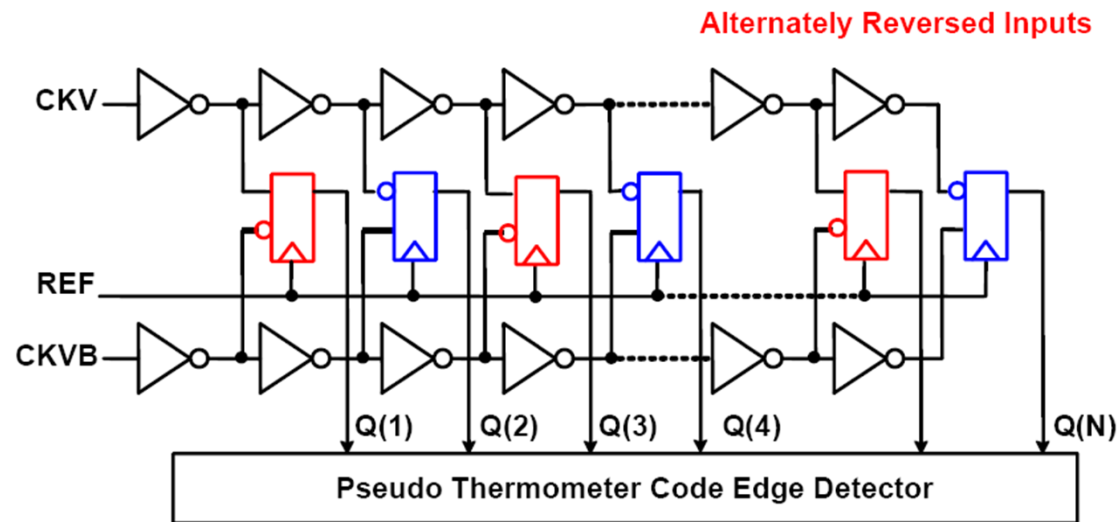
- Simple, but PLL dynamics is hard to be predicted.

- Linear – Conventional TDC

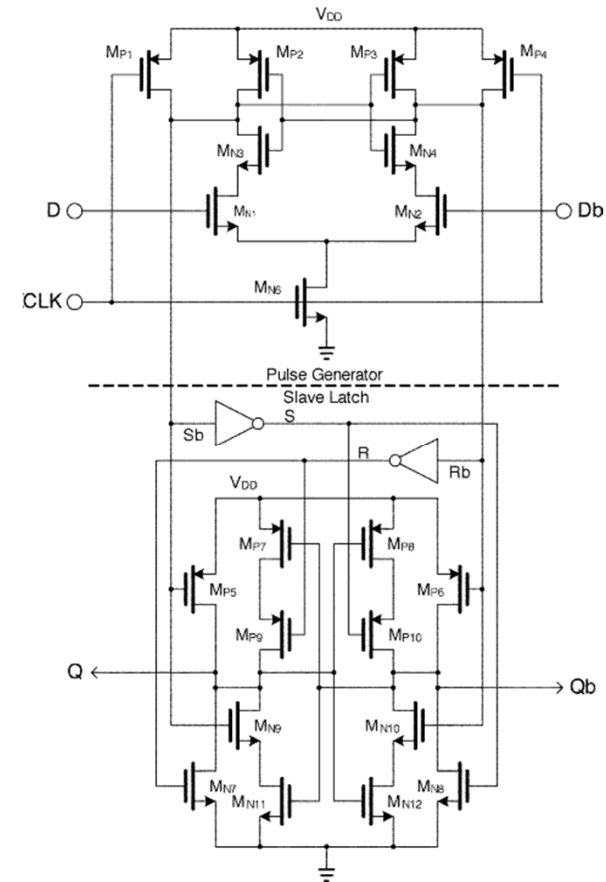


- Measure the time difference between CKV rising edge and REF rising edge
- Resolution = Two inverter delay (~ 40ps in 90nm CMOS)

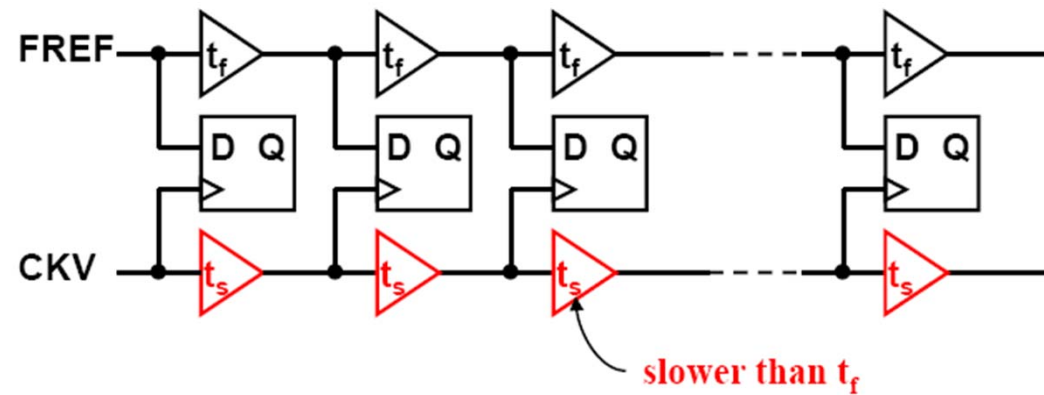
- Resolution improvement #1 – alternately reversed sampling



- Resolution = 1 inverter delay ($\sim 20\text{ps}$)
- Symmetric DFF is required which has same setup time for '1' and '0'.
- CKV and CKVB delay chain must be matched.

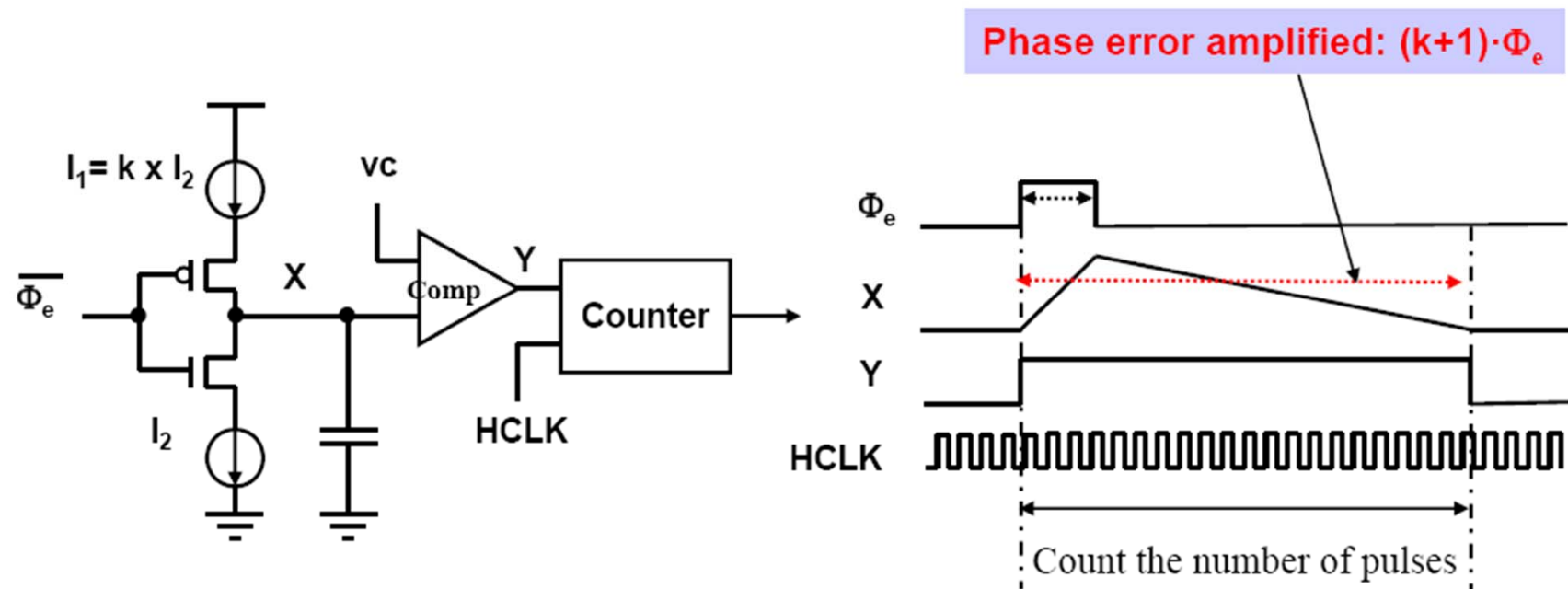


- Resolution improvement #2 – vernier delay line



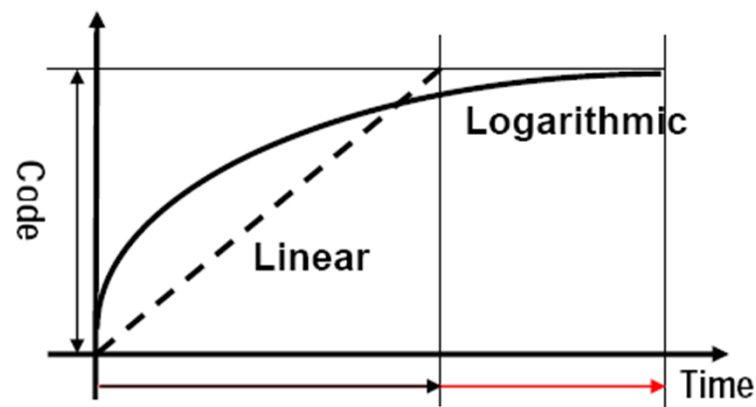
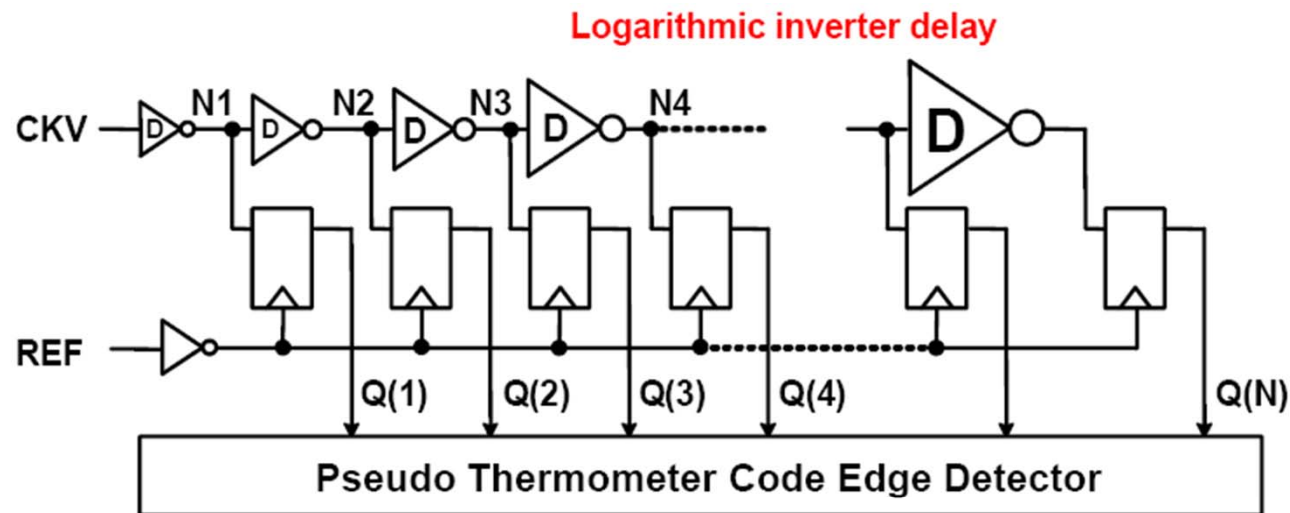
- Resolution = $t_s - t_f$
- Very high resolution (<1ps) can be achieved.

- Resolution improvement #3 – time amplifying



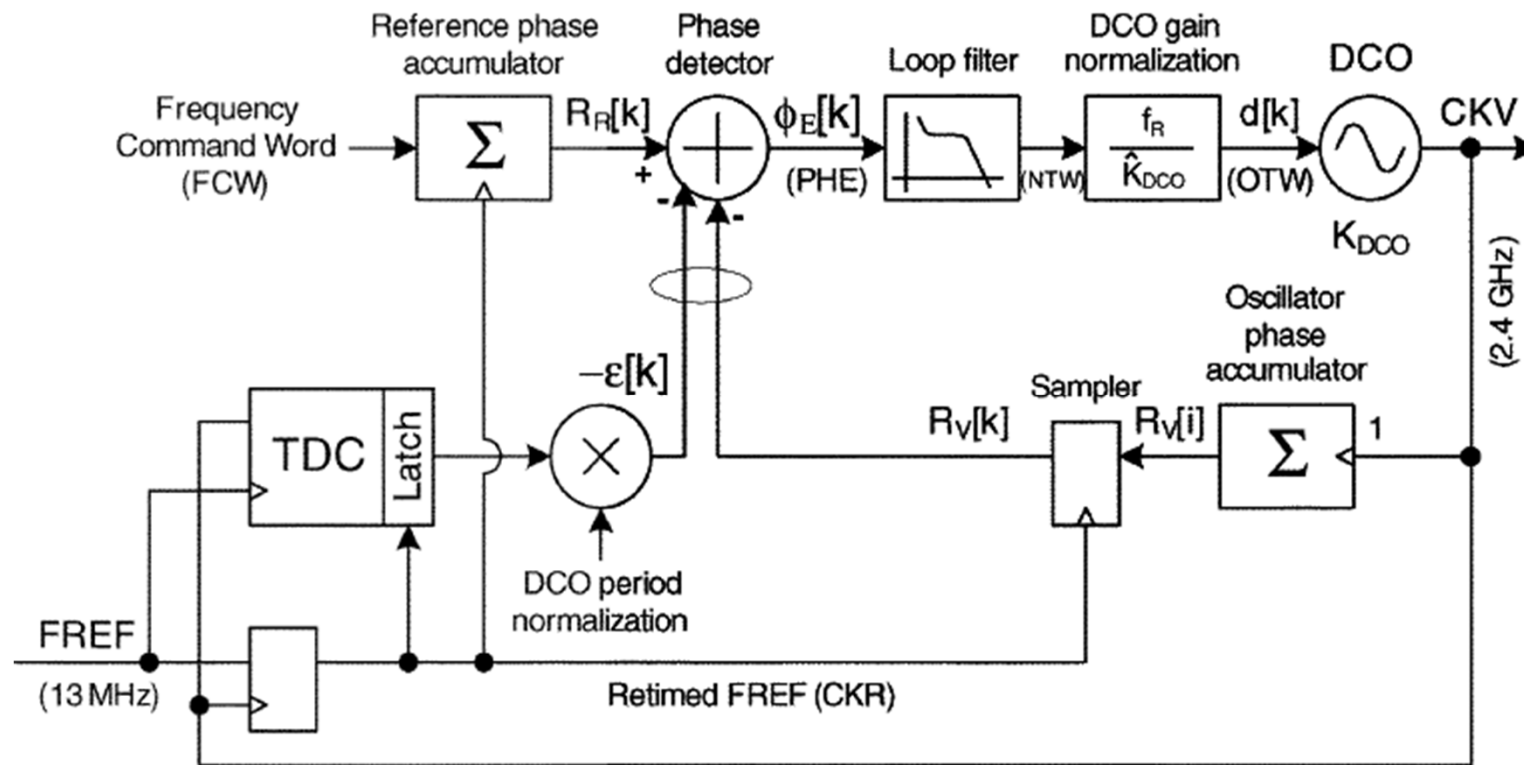
- Very high resolution can be achieved.

- Logarithmic TDC



- Represented range is extended
- Additional logic is required for linearization

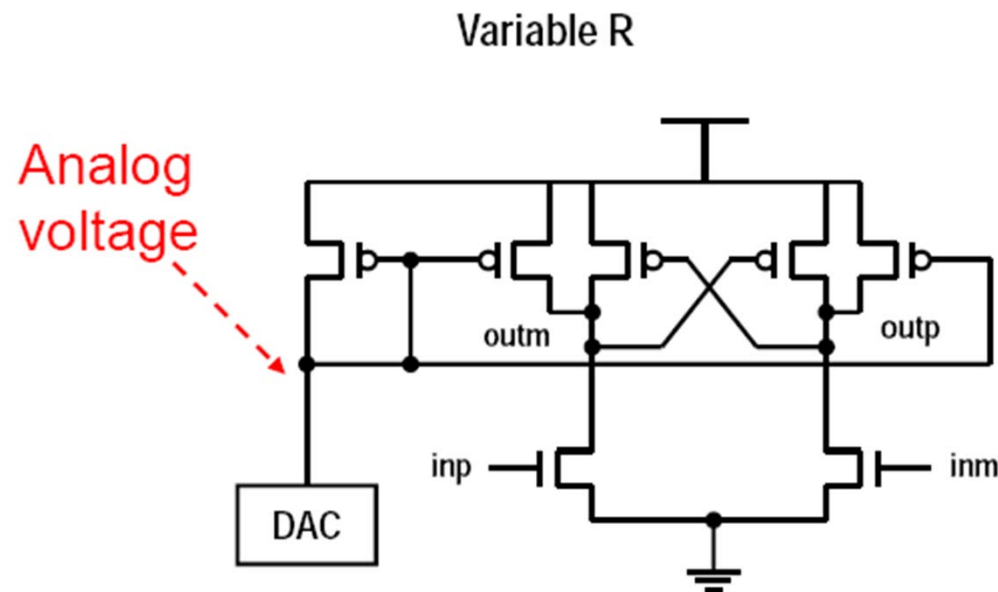
- Counter-aided TDC



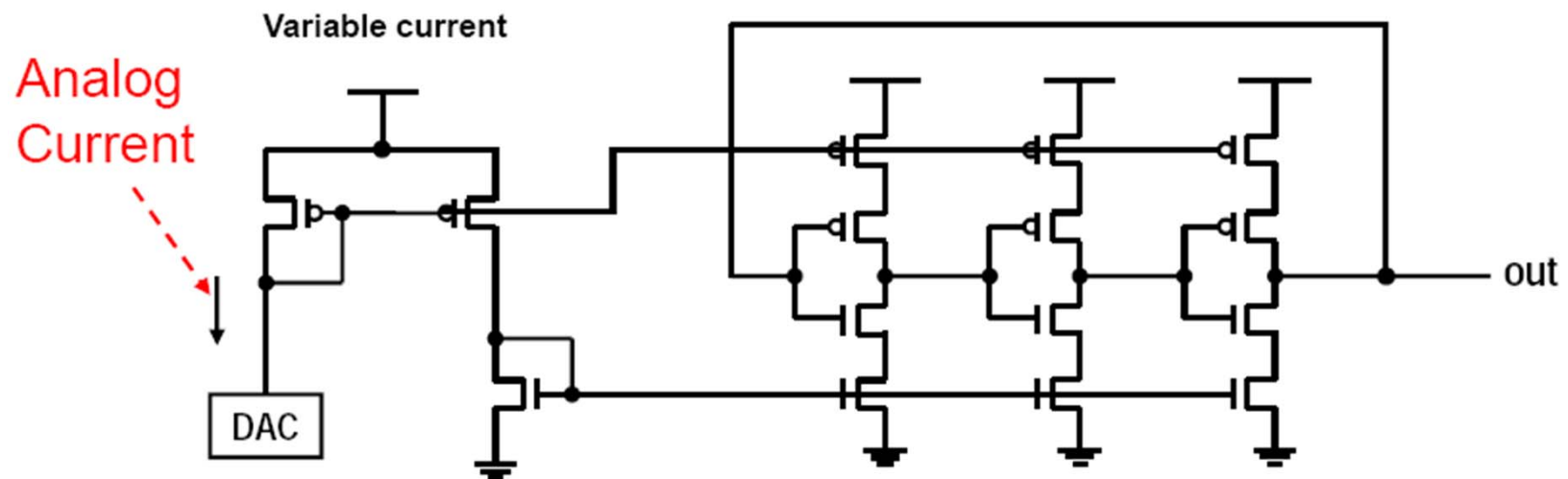
- TDC range can be expended by large-bit counter.
- High-speed counter is required.

- Digitally controlled oscillator
 - Analog approach (explicit DAC)
 - DAC + VCO
 - DAC + CCO
 - Digital approach (generic DAC)
 - Number of inverter stages
 - Number of drivers (variable inverter strength)
 - Digitally controlled varactor (segmented LC oscillator)
 - High freq. oscillator + programmable divider

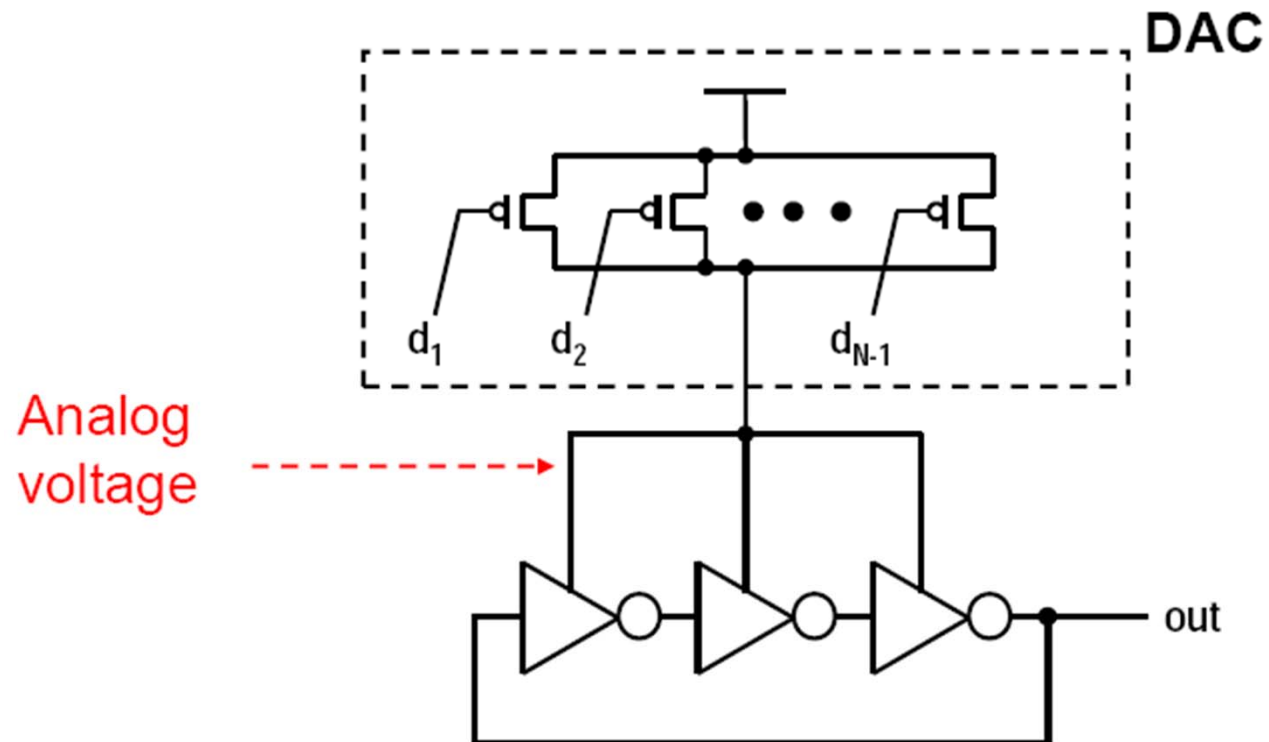
- Digitally controlled oscillator #1 – analog approach
 - DAC + analog VCO



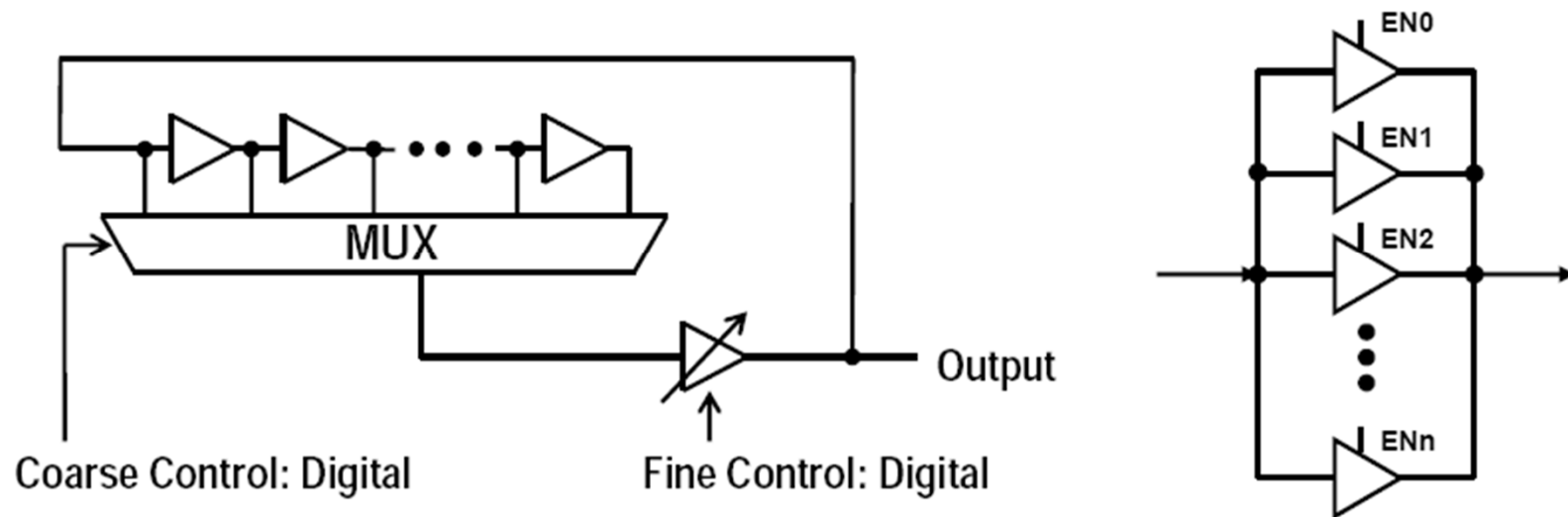
- Digitally controlled oscillator #2 – analog approach
 - DAC + analog CCO



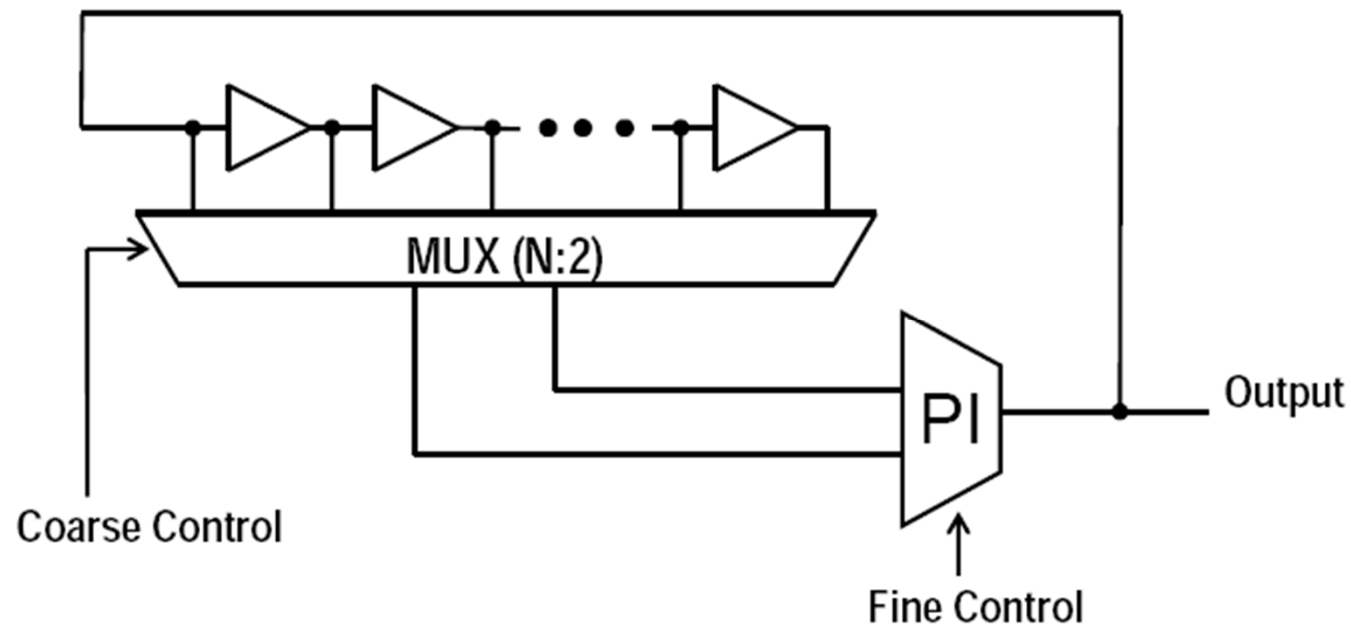
- Digitally controlled oscillator #3 – analog approach
 - Supply voltage control



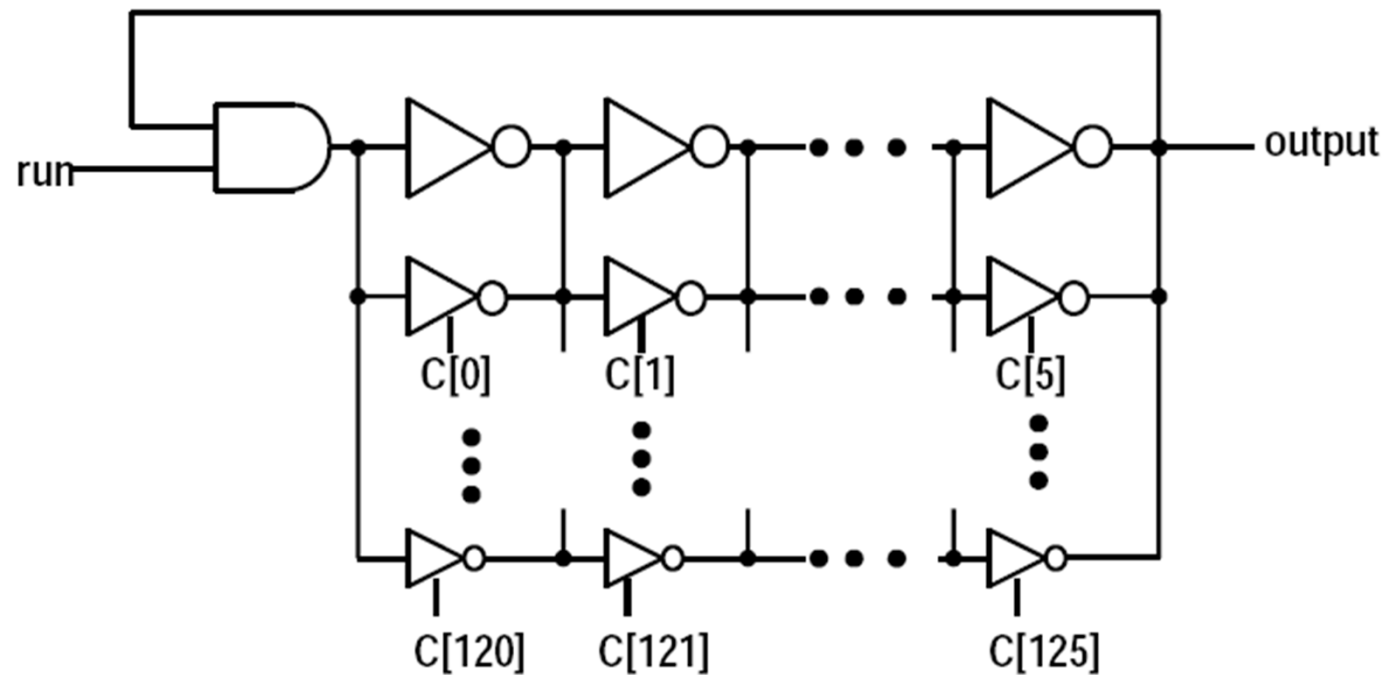
- Digitally controlled oscillator #4 – digital approach
 - Coarse: controls the number of delay elements
 - Fine: controls the driver strength



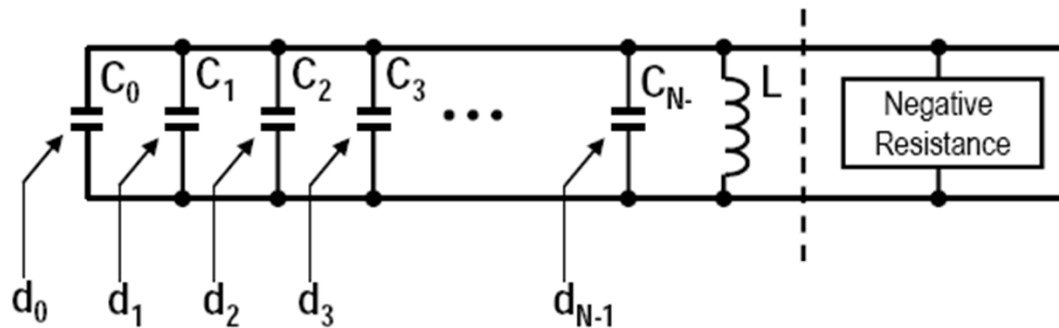
- Digitally controlled oscillator #5 – digital approach
 - Coarse: controls the number of delay elements
 - Fine: phase interpolation (PI)



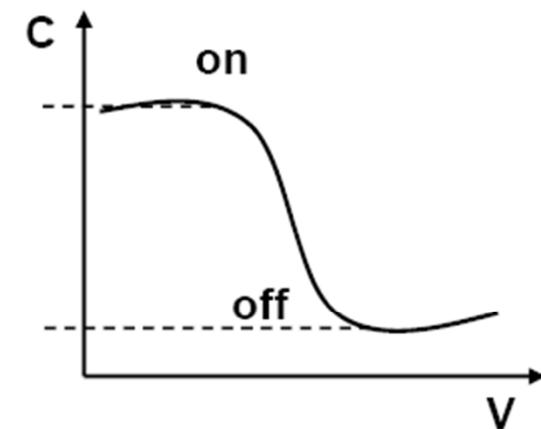
- Digitally controlled oscillator #6 – digital approach
 - Change inverter strength by varying the number of drivers based on digital control word



- Digitally controlled oscillator #7 – digital approach
 - Segmented LC-VCO
 - Change freq. by turning on/off binary- or equally-weighted small capacitances



$$\omega_0 = \frac{1}{\sqrt{LC}}$$



C-V curve of
Deep submicron
CMOS process